6.7 Modeling with Exponential and Logarithmic Functions

Name _____ Period ___ Date _____

Algebra II 6.7 (30-minute plan) · Student sheet

I can…

- tell whether a table shows a linear function, an exponential function, or neither.
- write an exponential function through two points.
- use technology to find an exponential model and make predictions.

Vocabulary · fill in the missing word

Constant differences — equal _____ and equal differences in y: linear

Constant ratios — equal x-steps and equal ratios of y: _____

Exponential regression — a _____ or Desmos model $y = ab^x$ that best fits data

Part 1 · Notes and examples (fill in each note, then work the example)

Linear or exponential?: Constant _____ → linear. Constant _____ → exponential.

Example 1

Is the table linear, exponential, or neither?

x	0	1	2	3	4
y	5	10	20	40	80

Two points: To find $y = ab^x$ through two points: substitute both points, then _____ one equation by the other to find b.

Example 2

Write an exponential function $y = ab^x$ whose graph passes through (1, 6) and (3, 54).

Exponential regression: In Desmos, exponential regression is typed as _____.

Example 3

A town's population (in hundreds) t years after 2020 is shown. Use exponential regression to find a model. Predict the population in 2026.

t	0	1	2	3	4
y	120	150	185	232	290

Part 2 · Partner practice

Partner practice

Is the table linear, exponential, or neither? If exponential, write the function.

x:	0	1	2	3
y:	2	6	18	54